

Venkat Terikuti

Senior Machine Learning Engineer | Data Scientist
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Professional Summary

Senior Machine Learning Engineer with strong backend engineering with 7+ years of experience building and productionizing ML systems across fintech, banking, telematics, and e-commerce. Hands-on expertise with deep learning and transformer-based models, generative AI/RAG services, and predictive analytics (fraud, churn, demand forecasting, anomaly detection), combined with robust MLOps on Azure/AWS using Docker, Kubernetes, and CI/CD. Proven track record designing feature pipelines and data platforms (Databricks, Snowflake, Azure Synapse), exposing low-latency ML APIs, and implementing monitoring, drift detection, and automated rollback to keep models reliable in production. Known for partnering closely with product, risk, and CX teams to translate business goals into experiments and AI features that drive measurable impact on self-service, onboarding time, conversion, and risk metrics, while mentoring junior engineers on scalable, reproducible ML practices.

EXPERIENCE

Senior ML Engineer

May 2025 – Present

Client: Fiserv

New Jersey

- Worked as a senior ML + backend engineer on Fiserv's client servicing and merchant experience platforms, contributing to AI features aimed at improving customer self-service, first-contact resolution, and merchant onboarding time, and wiring those outcomes into clear product KPIs.
- Designed and implemented generative AI / RAG services for payment transactions, disputes, and cardholder inquiries: built ingestion and chunking pipelines for transaction logs and support histories, created embedding indexes, and exposed LLM-backed APIs that returned context-aware responses with bounded latency and cost.
- Developed and productionized ML models for payment fraud, MCC classification, and transaction anomaly detection using transformer-based architectures and streaming features; deployed these as low-latency scoring services that improved detection latency and reduced false positives by ~25%.
- Helped architect end-to-end MLOps and backend delivery pipelines across Fiserv's multi-tenant payments platforms—using Azure DevOps, Git, Docker, Kubernetes, and MLflow (or equivalent) for model versioning, canary/shadow releases, drift monitoring, and automated rollback of underperforming models.
- Implemented and integrated REST/gRPC APIs with Fiserv Developer Studio and internal ledger systems for merchant onboarding, push payments, and card issuer flows—defining schema contracts, versioning, and telemetry, and contributing to an observed ~30% reduction in integration defects over subsequent releases.
- Partnered with product, risk, compliance, and CX teams to translate requirements into AI features and experiments (e.g., conversational onboarding/KYC assistant), instrumented dashboards for model and business KPIs (self-serve rate, backlog, fraud loss), and supported A/B and champion–challenger tests that drove measurable improvements such as ~20% reduction in average merchant onboarding time.

ML Research Assistant (Post Grad research)

Aug 2024 – Apr 2025

University of Arizona

Tucson, AZ

- Built a scalable semantic search engine for e-commerce by fine-tuning Sentence Transformers and Google FLAN-T5 on a 1.3M-product catalog, integrating FAISS for fast vector retrieval and enabling real-time, intent-aware search.
- Implemented a full-stack ML pipeline covering data prep, model fine-tuning, and dynamic filter extraction (e.g., price, ratings), enhancing structured query refinement and delivering a 90% increase in Precision@10 over traditional keyword search.
- Led applied research on LLM-powered search, authored a peer-reviewed paper titled "*LLM-based Semantic Search for Conversational Queries in E-commerce*" submitted to CIKM'25 (ACM), showcasing innovative use of Sentence-BERT for user query understanding and product discovery for public sector use cases.

Senior Software Engineer – Machine Learning

Nov 2021 – Jan 2023

Client: Hitachi GlobalLogic

Hyderabad, India

- Designed and deployed deep learning demand-forecasting pipelines using TensorFlow/Keras and attention-based LSTMs for 100,000+ SKUs, achieving ~96% accuracy on hold-out sets and reducing forecast error by ~35% versus legacy statistical methods.
- Owned the end-to-end MLOps lifecycle on AWS SageMaker—training, hyperparameter tuning, batch jobs, and real-time endpoints—for 10+ production models; integrated CI/CD with Jenkins/GitLab CI, Docker, and Kubernetes to cut deployment latency by ~40% and standardize rollouts across staging and prod.
- Architected data ingestion, cleansing, and feature-engineering workflows in MySQL, Hive, Airflow, and Databricks to process terabytes of clickstream and transactional data, improving data reliability and enabling cross-team analytics and model reuse.
- Built and productionized a real-time anomaly detection pipeline using Python and Kafka streaming to monitor order volumes, app performance, and fulfillment bottlenecks, enabling proactive incident detection and reducing mean time to resolution by ~30%.
- Partnered with marketing, supply-chain, operations, and inventory teams to translate business problems into ML solutions

(campaign uplift, supply-chain repositioning, SKU-level inventory optimization), and validated impact through A/B tests, multivariate experiments, user-segment analysis, and decision-maker facing dashboards.

- Led model governance and team best practices—implementing model versioning, monitoring and drift detection, automated retraining and rollback strategies, as well as mentoring data scientists/ML engineers on scalability and latency optimization—while advising leadership on ML roadmap, tooling choices, and cost efficiency (e.g., instance/right-sizing, model quantization).

Senior Data Scientist

Nov 2019 – Oct 2021

Client: Lytx

Remote

- Led a globally distributed ML & analytics team (US, Argentina, Poland) building telematics-driven safety and risk models on millions of daily events (video, sensor, GPS), delivering actionable driver-risk scores and fleet performance insights for a mission-critical platform.
- Architected and optimized high-throughput feature pipelines in Databricks Delta Lake, Snowflake, and SQL to ingest 6M+ telematics events per day with >99.9% data freshness, enabling real-time modelling of driver behavior, risky events, and fleet operations at scale.
- Defined a robust analytics & ML foundation by establishing a dbt-based semantic layer, dimensional models, and reusable data marts, unlocking self-service experimentation (A/B tests, cohorts) for product, safety, operations, and compliance teams and cutting reporting turnaround by ~40%.
- Worked hands-on across the stack—designing AWS Glue / Airflow workflows, tuning Spark/Databricks jobs, optimizing queries, and collaborating with SRE/DevOps—to ensure reliable model pipelines, low-latency scoring, and highly available data services for global customers.
- Partnered with product, safety, and behavioral science stakeholders to design and analyse 10+ experiments (e.g., driver coaching, risk alert policies), and mentored junior data scientists/engineers on telematics feature engineering, anomaly detection, behavioral scoring, and reproducible ML best practices aligned with Lytx's MV+AI fleet safety vision.

Data Scientist

Mar 2016 – Oct 2019

Geetham Software

Chennai

- Led ML-driven customer analytics for the bank's Azure ecosystem—building and maintaining ETL/feature pipelines across marketing, finance-ledger, and CRM systems, and introducing standardized data quality checks, lineage tracking, and SLA dashboards to keep models and BI “audit ready.”
- Designed and deployed segmentation, churn prediction, and propensity scoring models powering inbound/outbound CRM campaigns; worked with business stakeholders to define KPIs, and packaged quarterly business review insights for executive leadership and product decision-making.
- Owned production reliability for model and data pipelines by implementing monitoring, automated failure-handling, and RCA playbooks, and by enforcing upstream data contracts with source-system teams—reducing recurring pipeline/model breaks by ~30%.
- Drove cross-functional adoption of ML by collaborating with product managers and CRM leads to frame testable hypotheses, run cohort/holdout experiments, and measure uplift/ROI for key journeys, while mentoring junior data scientists/engineers on feature engineering, documentation, and reproducible ML workflows in Azure Synapse/Data Lake.

EDUCATION

Master of Science in Data Science, The University of Arizona (GPA: 3.89)

Dec 2024

Bachelor of Technology in Mechanical Engineering, Acharya Nagarjuna University (GPA: 3.42)

May 2015

PROJECTS

Building LLM From Scratch.

- Implemented GPT-style large language model from ground up using PyTorch with byte-level tokenizer and scaled dot-product attention. Built complete training pipelines including pretraining on unlabeled text, classification finetuning, and instruction finetuning (SFT). Demonstrated deep understanding of attention mechanisms, gradient stabilization, and modern LLM architectures.

High-Performance Video Diffusion Model Optimization – Wan 2.2

- Optimized the Wan 2.2 text-to-video diffusion model pipeline for H100 GPUs, leveraging advanced context-parallelism, flash attention 3, and direct cuBLAS/triton kernel integration to achieve sub-25s 720p video generation for 81 frames—outperforming standard FA2/FA3 baselines (15–30 TFLOPS, 2–4× speedup).

Skills

Languages: Python, SQL, Bash, C++

Machine Learning & AI: Deep Learning (TensorFlow, Keras), Transformer Models, LLMs & RAG (Retrieval-Augmented Generation), Time-Series Forecasting, Anomaly Detection, Fraud Detection, Churn & Propensity Modeling, Feature Engineering, Model Explainability, Recommender Systems

Data Engineering & Big Data: Apache Spark, PySpark, Databricks, Delta Lake, Kafka, Airflow, AWS Glue, ETL/ELT Pipelines, Dimensional & Star/Snowflake Data Modeling, Data Warehousing, Data Quality & Lineage, dbt Core

Cloud & Data Platforms: AWS (SageMaker, Lambda, Glue, S3, EC2), Azure (Data Factory, Data Lake, Synapse, Azure DevOps), Snowflake, Hadoop Ecosystem, SQL Databases (MySQL, Oracle, SQL Server), BI Tools (Tableau, Power BI)

MLOps, Backend & DevOps: MLflow (or equivalent), Model Versioning & Deployment, CI/CD (GitLab CI, Jenkins, Azure DevOps), Docker, Kubernetes, REST/gRPC Services, Monitoring & Observability (Prometheus, Grafana), Testing & Release Strategies (canary/shadow, rollback)

Analytics, Experimentation & Leadership: A/B Testing & Multivariate Experiments, Cohort Analysis, KPI Definition & Dashboards, Agile/Scrum, Cross-Functional Collaboration (Product, Risk, CX, Marketing), Mentorship, Stakeholder Management, Change Management, OKR-Driven Delivery

Certifications

Deep Learning Specialization – Andrew Ng, Coursera
SQL(Advanced) - Hackerrank